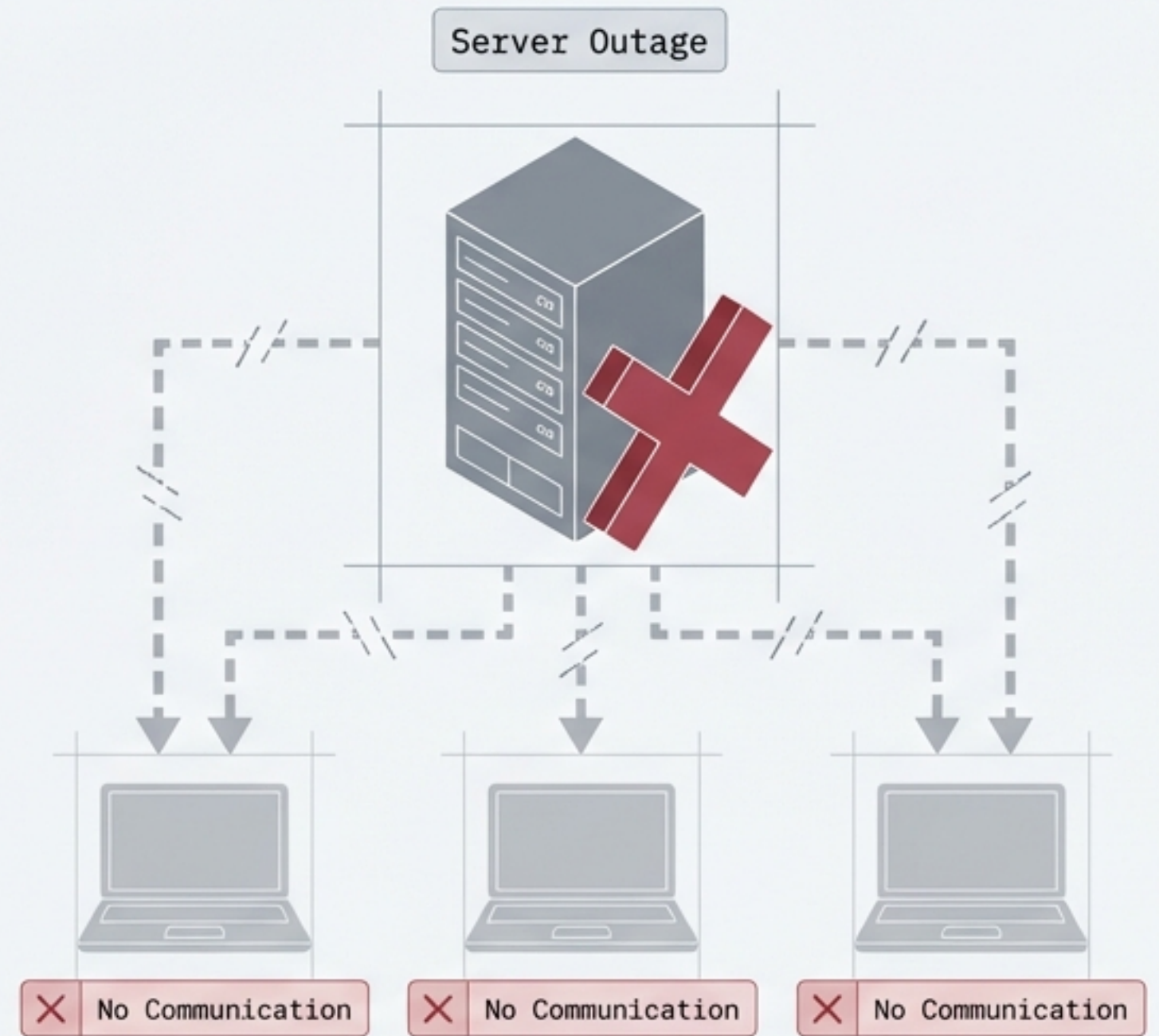
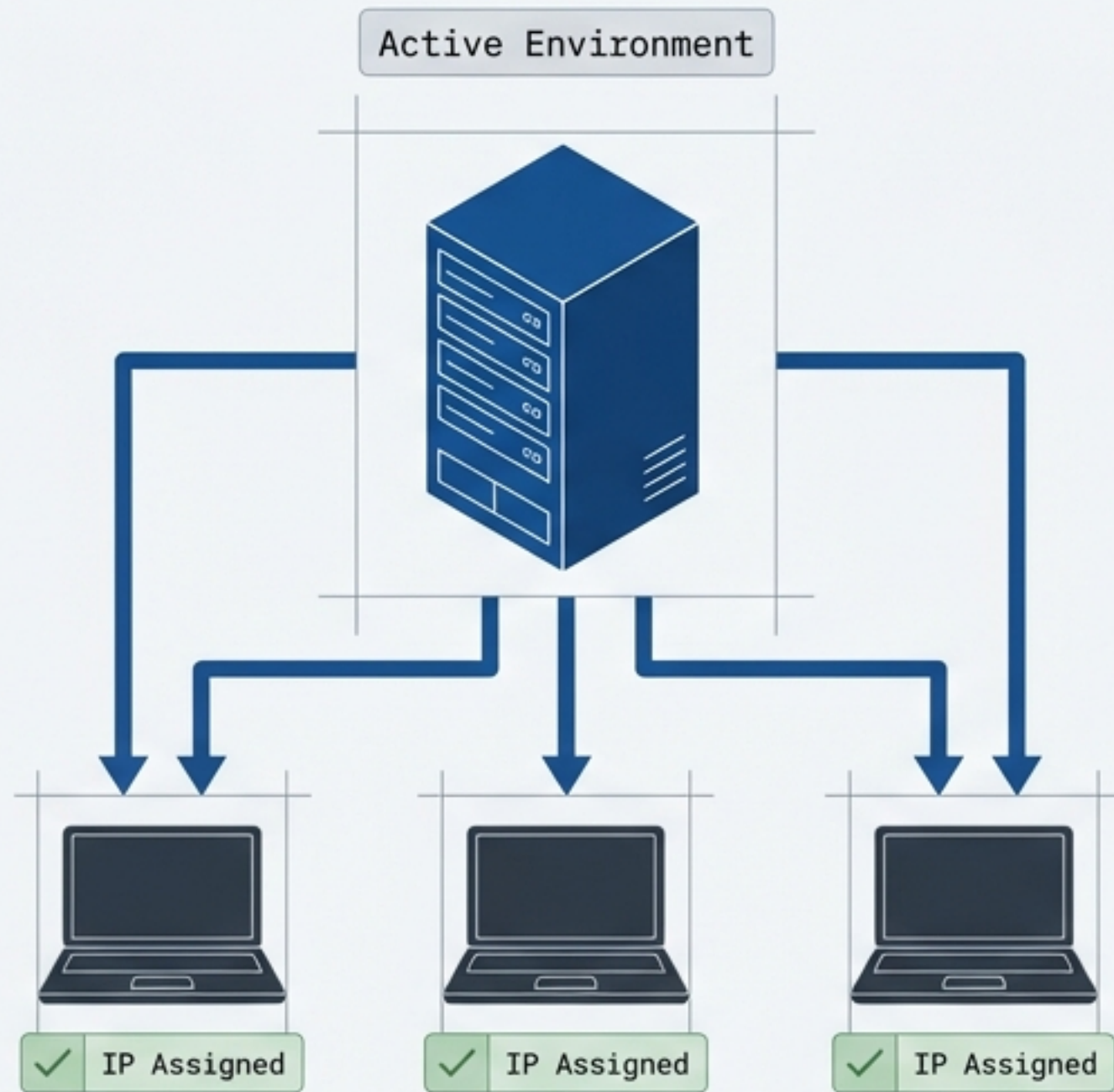


Zero Downtime DHCP

The ultimate step-by-step guide to configuring High Availability, Load Balancing, and Fault Tolerance for your IP infrastructure in Windows Server 2019.



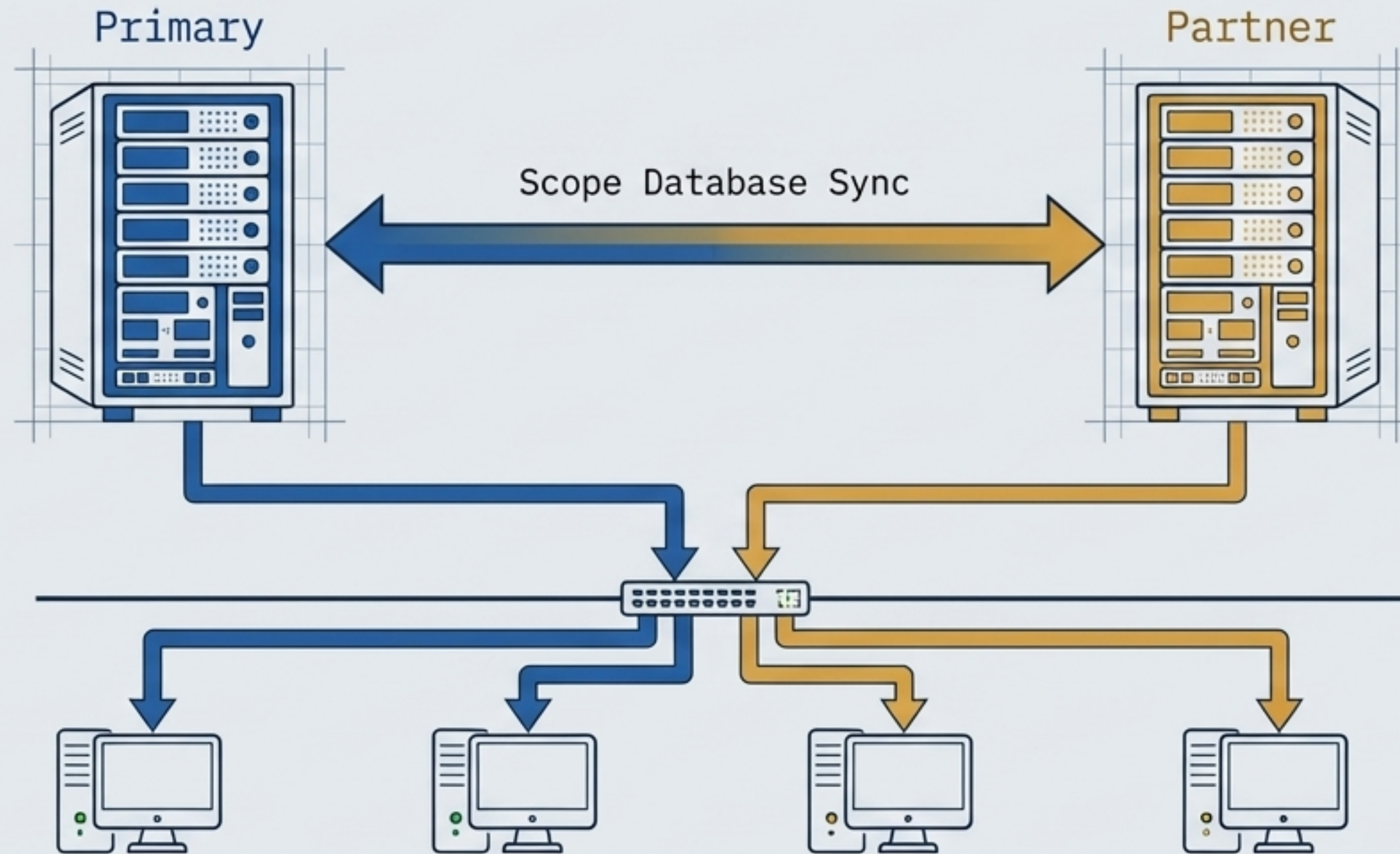
The Single Point of Failure



- Without DHCP, devices cannot obtain IP addresses dynamically.

- A single server outage blocks all network communication, preventing users from accessing essential network resources and services.

Continuous IP Assignment Through Failover



DHCP Failover is a native feature that replicates DHCP scope databases across two distinct servers.

If the primary server goes completely offline, the partner server instantly assumes responsibility, ensuring seamless IP assignment and zero network disruption.

The Core Pillars of a Resilient Network



High Availability

Ensures continuous, 24/7 uptime for critical network IP assignment services.



Load Balancing

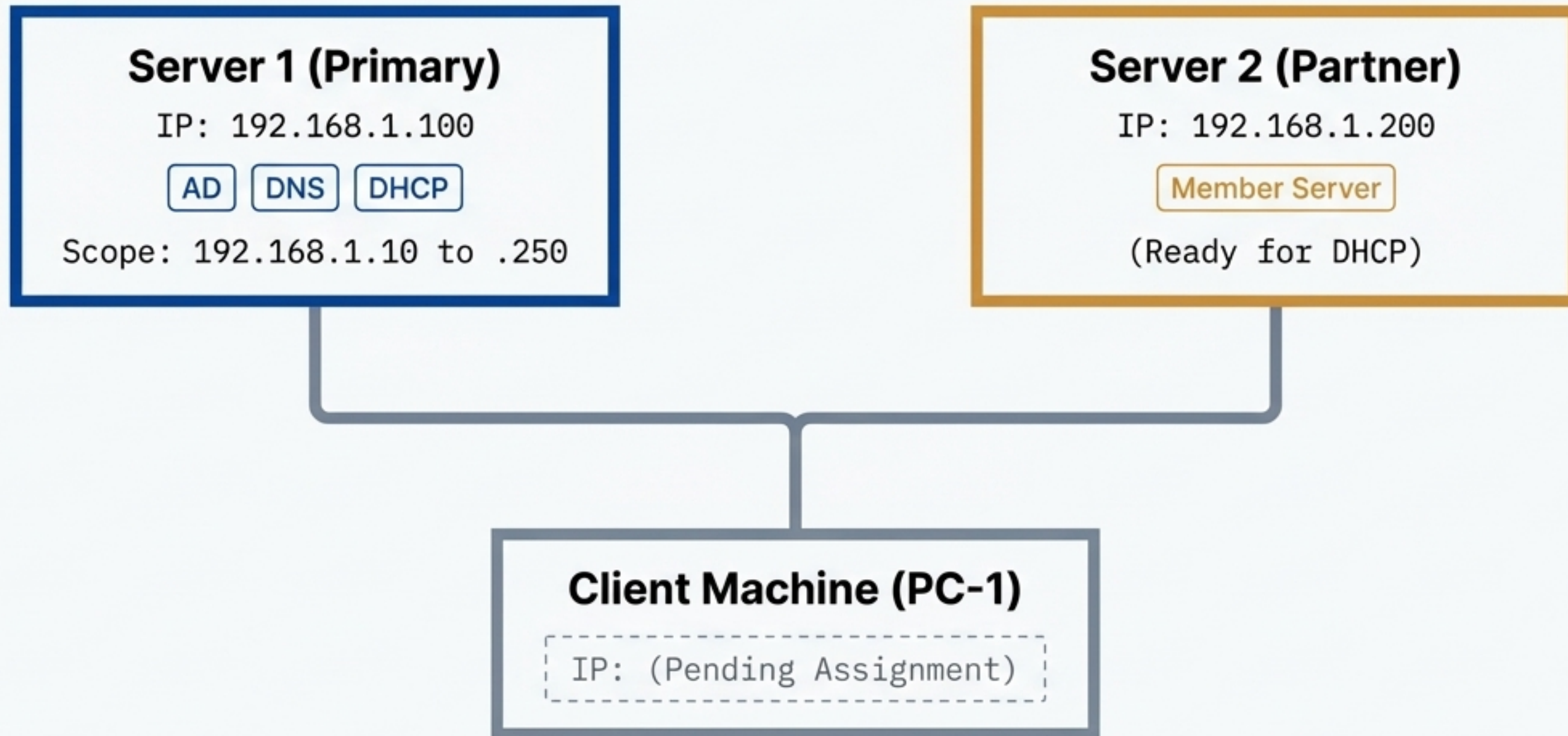
Actively distributes client IP leasing requests across multiple servers to prevent bottlenecks.



Fault Tolerance

Guarantees automatic, seamless transition and recovery during unexpected hardware or service failures.

The Lab Environment Blueprint

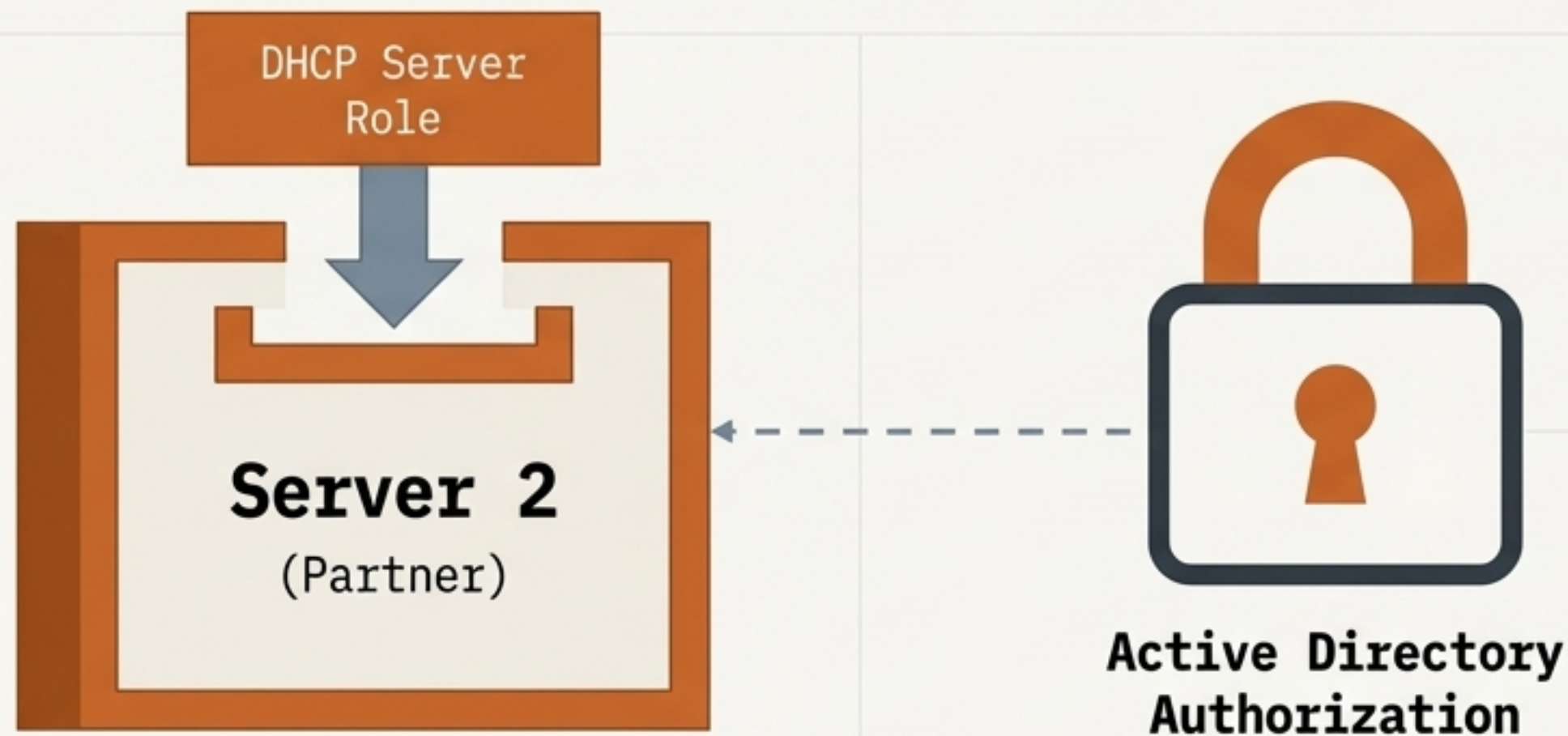


This baseline topology outlines the required infrastructure before initiating the failover configuration. Server 1 holds the active DHCP scope, while Server 2 is a blank slate.

Diagnostic Matrix: Choosing Your Operational Mode

	Load Balance Mode	Hot Standby Mode
Architecture	Active-Active Both servers issue IPs simultaneously.	Active-Passive Partner waits until Primary fails.
Default Split	50% / 50% Customizable distribution.	Primary handles 100% Partner reserves 5% for emergencies.
Best For	Single-site networks with high traffic volumes.	Multi-site or branch office deployments with a centralized backup.

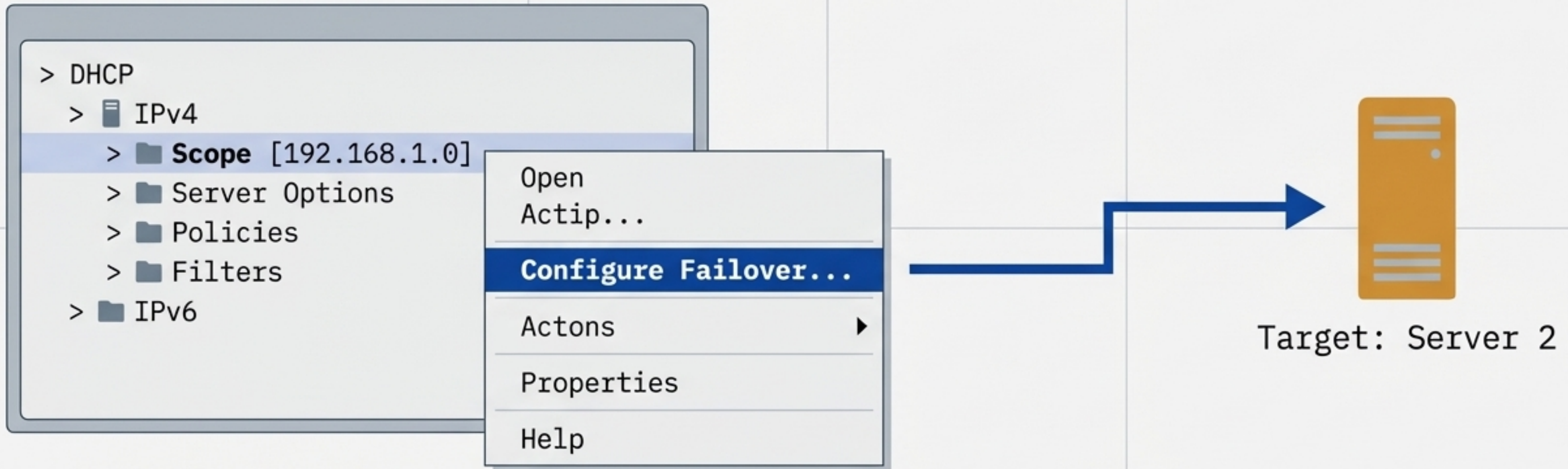
Preparing the Partner Server



Install the DHCP Server Role on the designated Member Server (Server 2).

Critical Requirement: You must explicitly Authorize Server 2 in Active Directory. This creates the necessary DHCP Administrators and Users security groups, making the server functional on the domain.

Initiating the Failover Setup



- Failover is configured per scope, not globally.
- From the primary server (Server 1), right-click the target IPv4 scope and launch the configuration wizard to designate Server 2 as the partner server.

Defining the Failover Relationship

Srv1-Srv2-192.168.1.0

Relationship Name: Standardized naming convention for easy identification.

Load Balance

Mode Selection: Operational mode and distribution split.

50%

50%



Shared Secret: Secure authentication key for server-to-server communication.

Synchronizing the Database

Server 1 Data

Pool:	192.168.1.10 - .250
Exclusions:	192.168.1.50 - .60
MAC Res:	00:1A:2B:3C:4D:5E -> 192.168.1.100
MAC Res:	00:1A:2B:3C:4D:5F -> 192.168.1.101
Scope Options:	Router 192.168.1.1, DNS 8.8.8.8

**INSTANT
REPLICATION**

Server 2 Data

Pool:	192.168.1.10 - .250
Exclusions:	192.168.1.50 - .60
MAC Res:	00:1A:2B:3C:4D:5E -> 192.168.1.100
MAC Res:	00:1A:2B:3C:4D:5F -> 192.168.1.101
Scope Options:	Router 192.168.1.1, DNS 8.8.8.8

Refresh to view

Upon completion, the entire DHCP scope—including IP address pools, exclusions, and active MAC reservations—instantly replicates across the network to the partner server.

Establishing the Baseline: Client Perspective

```
C:\> ipconfig /all
```

Windows IP Configuration

Ethernet adapter Ethernet:

```
Connection-specific DNS Suffix  . : yourdomain.local
Link-local IPv6 Address . . . . . : fe80::f44d:5b98:567f:6589%12
IPv4 Address. . . . . : 192.168.1.15
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
DHCP Server . . . . . : 192.168.1.100
DNS Servers . . . . . : 192.168.1.1
NetBIOS over Tcpip. . . . . : Enabled
```

Lease sourced
from Server 1

Before simulating a failure, verify normal operations. The client successfully requests and receives its initial IP lease directly from the primary server (Server 1).

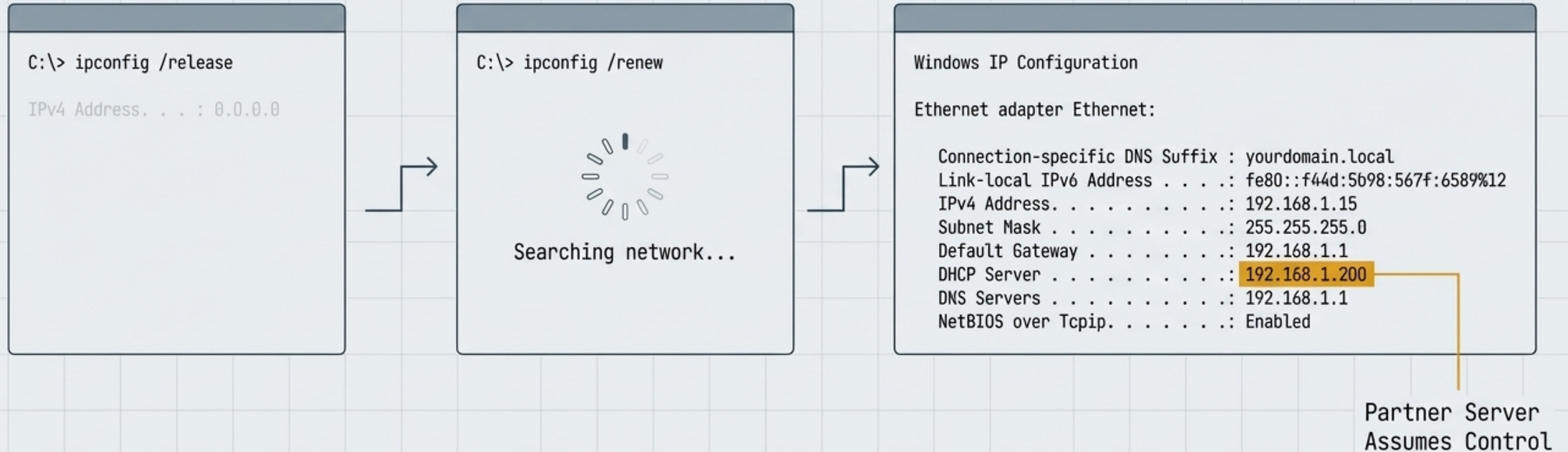
Simulating a Primary Server Disaster



To prove absolute fault tolerance, we intentionally sabotage the environment.

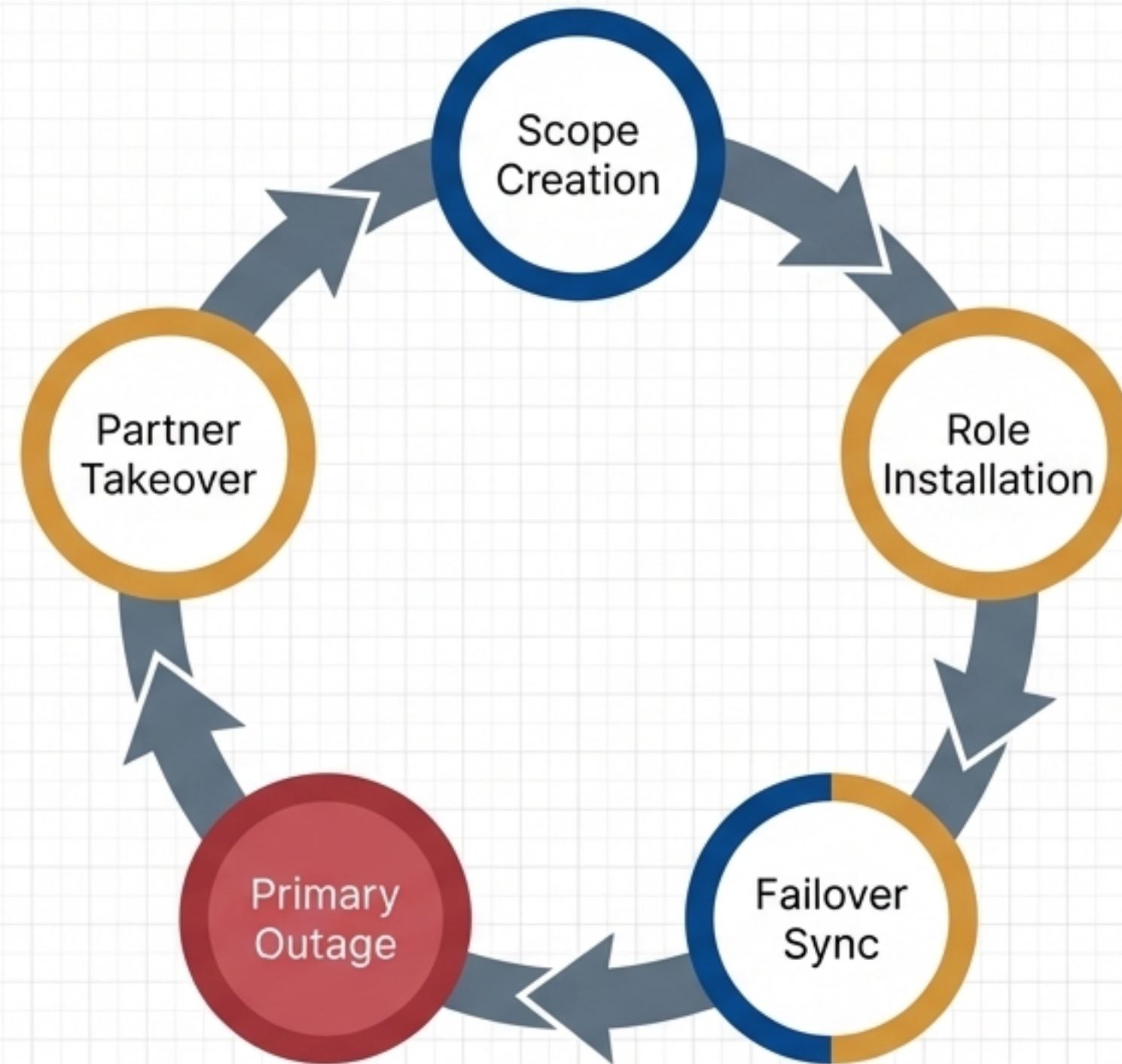
By manually stopping the DHCP Server service on Server 1 via the management console, the primary IP lease provider is forced completely offline.

The Seamless Handover



"Server 2 detects the primary outage and instantly steps into the void. It processes the renewal request and maintains the lease. The network remains fully operational."

The Failover Lifecycle



Resiliency requires intentional architecture. DHCP failover transforms a fragile, single-point-of-failure network into a self-healing infrastructure where hardware crashes never cascade into total blackouts.

Deployment Checklist

	Deploy redundant servers and install the DHCP role.
	Explicitly Authorize the partner server in Active Directory .
	Choose between Load Balancing (Active-Active) or Hot Standby (Active-Passive).
	Configure the failover relationship, verify database replication, and simulate an outage to test recovery.

Secure your network by eliminating the DHCP single point of failure today.